

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a)      | 1 number of / range of (different), ecosystems / habitats ;<br>2 number of species and their relative abundance ;                                                                                                                                                                                                                                                                                                                                                                               | <b>2</b> |
| 2(b)      | any <b>three</b> from:<br><br>1 to, store / contain, multiple / many, genome / DNA / nucleotide, sequences ;<br>2 to, align / compare / search, sequences ;<br>3 to, process / analyse / sort / calculate from, large quantities of data ;<br>4 to, share / access, data / information from, others / elsewhere ;<br>5 to save time / fast(er) ;                                                                                                                                                | <b>3</b> |
| 2(c)(i)   | <i>Anopheles</i> ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>1</b> |
| 2(c)(ii)  | three / 3 ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>1</b> |
| 2(d)(i)   | any <b>three</b> from:<br><br>1 less able / not able, to, adapt / evolve ;<br>2 little variation for selection to act on ;<br>3 few(er) (potentially), useful / beneficial, alleles ;<br>4 all / most, (could be) killed by same, disease / selection pressure ;<br>5 if due to small population size this decreases species survival chance ;                                                                                                                                                  | <b>3</b> |
| 2(d)(ii)  | low(er) genetic variation means high(er), conservation status / threat / vulnerability / endangerment / risk of extinction ;                                                                                                                                                                                                                                                                                                                                                                    | <b>1</b> |
| 2(d)(iii) | any <b>three</b> from:<br><br><i>does not support because</i><br>1 chimpanzee has, most / highest, genetic variation but is, EN / endangered ;<br>2 minke whale has (joint), least / lowest, genetic variation but is, LC / least concern ;<br>3 genetic variation, similar / same, but conservation status is different ;<br>4 example ;<br>5 gorilla is, most / critically, endangered but has high(er) genetic variation ;<br>6 AVP ; e.g. <i>ref. to data limited to only seven species</i> | <b>3</b> |